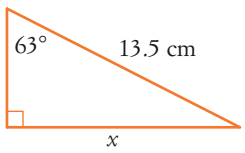
Name: _____

WORKSHEET

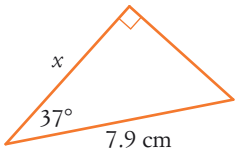
# Basic trigonometry

**1** Find the value of $x$ in these triangles correct to two decimal places.

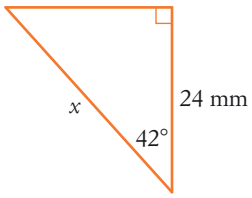
**a**



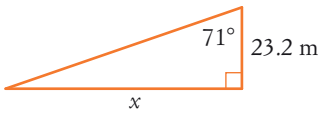

**b**



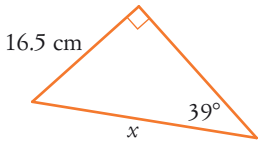

**c**



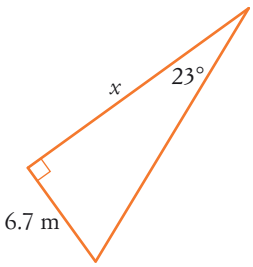

**d**




**e**



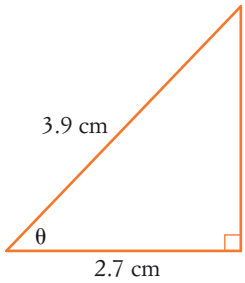

**f**



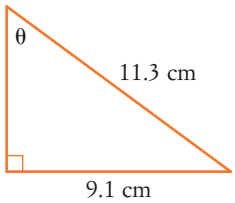

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**2** Find the value of θ in these triangles correct to one decimal place.

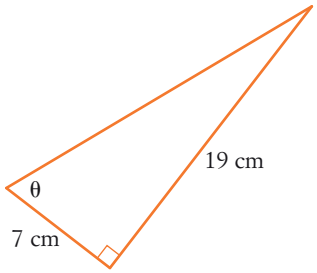
**a**



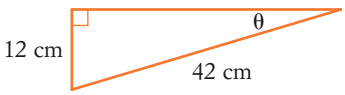

**b**



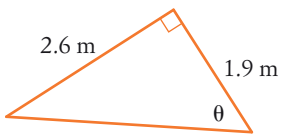

**c**



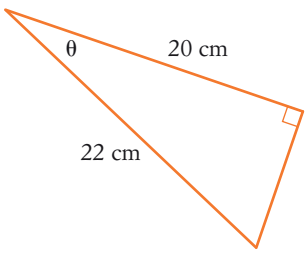

**d**




**e**




**f**




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**3** Identify the **i** period and **ii** amplitude for each of these functions.

**a** $y = 3 \sin(3x)$

**b** $y = \frac{1}{2} \cos(5x)$

**c** $y = 3 \tan(x)$

**d** $y = 5 \cos\left(\frac{x}{3}\right)$

**e** $y = 4 \tan(\pi x)$

**f** $y = \frac{1}{4} \sin\left(\frac{\pi x}{2}\right)$

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**4** Sketch the following graphs for the domains listed.

**a** $y = \tan(x)$ for $[0, 2\pi]$

**b** $y = \sin(2x)$ for $[0, 2\pi]$

**c** $y = 2\cos(3x)$ for $[0, 2\pi]$

**d** $y = \frac{1}{2}\sin(4x)$ for $[0, 2\pi]$

**e** $y = 3\cos(\pi x)$ for $[0, 6]$

**f** $y = \frac{1}{3}\tan\left(\frac{\pi x}{2}\right)$ for $[0, 6]$

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## Answers

**1** **a** 12.03 cm

**b** 6.31 cm

**c** 32.30 mm

**d** 67.38 m

**e** 26.22 cm

**f** 15.78 m

**2** **a** 46.2°

**b** 53.6°

**c** 69.8°

**d** 16.6°

**e** 53.8°

**f** 24.6°

**3** **a** **i** $\frac{2\pi}{3}$ **ii** 3

**b** **i** $\frac{2\pi}{5}$ **ii** $\frac{1}{2}$

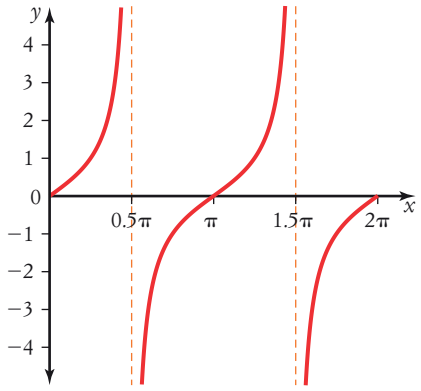
**c** **i** $\pi$ **ii** 3

**d** **i** $6\pi$ **ii** 5

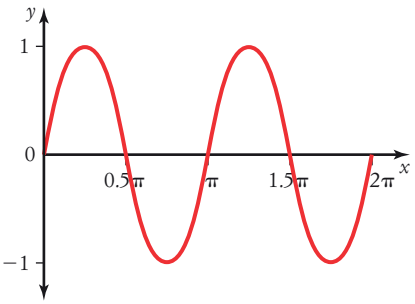
**e** **i** 2 **ii** 4

**f** **i** 4 **ii** $\frac{1}{4}$

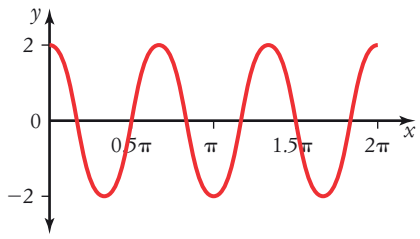
**4** **a**



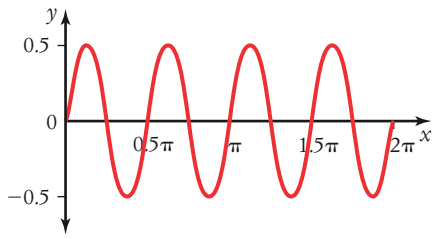

**b**



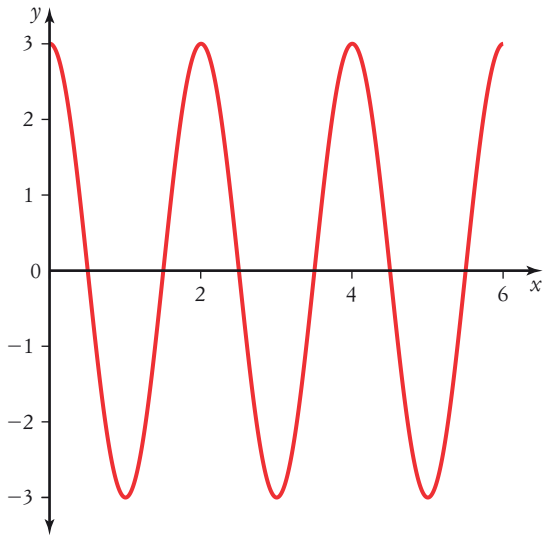

**c**



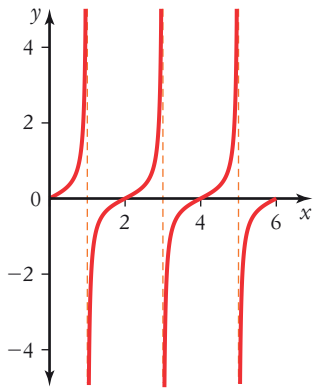

**d**




**e**




**f**




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